

AA7 - Properties of regular languages

(possíveis soluções – podem estar erradas)

NOTA: contacta-nos se tiveres as soluções corretas accio.feup.dei@gmail.com

1. When proving that a given language L is irregular with the pumping lemma, we have to show that there exists some word $w \in L$, with $|w| \geq n$ (where ' n ' defines the pumping length), and

- ☒ For every splitting $w=xyz$ in which $|xy| \leq n$ and $y \neq \epsilon$ there exists k such that $xy^kz \notin L$. I.e., we need to show that no decomposition works.
- ☐ Any splitting $w=xyz$ in which $|xy| \leq n$ and $y \neq \epsilon$ there exists k such that $xyz \notin L$. I.e., we only need to show that one decomposition does not work.

2. We cannot use the pumping lemma to prove that this language is not regular $\{ w w^p \mid w, p \in \{0, 1\}^+ \}$.

- ☐ True
- ☒ False

3. Which of the following regular expressions is the complement of the language represented by this regular expression?

$a(ba)^*(a + bb) + b(ab)^*(b + aa)$

- ☐ $a(ab)^*(a + ab) + b(ba)^*(b + ba)$
- ☒ $\epsilon + (\epsilon + b)a(ba)^* + (\epsilon + a)b(ab)^* + (a(ba)^*(a + bb) + b(ab)^*(b + aa))(a + b)(a + b)^*$
- ☐ $\epsilon + (\epsilon + b)a(ba)^* + (\epsilon + a)b(ab)^*$
- ☐ $(a(ba)^*(a + bb) + b(ab)^*(b + aa))(a + b)(a + b)^*$

4. Knowing that regular languages are closed to a given operation, when we apply such operation over an irregular language, we always obtain a regular language.

- ☐ True
- ☒ False

5. The intersection between two regular languages can result in an irregular language. 

- ☐ True
- ☒ False

6. We cannot build a DFA to represent all the words present in a "The Wall Street Journal" edition.

- ☐ True
- ☒ False

7. If we have a language L that we know to be irregular and a language M which we do not know its regularity, we can prove that M is not regular by

- ☐ applying closure properties over L to obtain M.
- ☒ applying closure properties over M to obtain L.

8. The closure of a regular language is always an infinite language.

- ☐ True
- ☒ False

9. The reverse of a regular language always results in a regular language.

- ☒ True
- ☐ False

10. When building the distinguishable states table, we can only say that state A is distinguishable from state B if

NOTE: a represents any symbol of the alphabet

- ☐ for all $\delta(A,a) = R$ and $\delta(B,a) = S$, R must always be distinguishable from S
- ☒ there is at least one $\delta(A,a) = R$ and $\delta(B,a) = S$, where R is distinguishable from S